De La Salle University
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System Administration Laboratory Manual

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Section:	S18	Grade:	

10.0 Public Key Authentication for SSH

Public Key Authentication (PKA) is an alternative way of securely logging in using SSH. Normally, a user will need to provide a set of credentials (username and password). When using PKA, a cryptographic key pair is required instead of the credentials. PKA provides additional cryptographic security when logging in remotely and allows for passwordless logins.

Some benefits of using PKA include:

- The cryptographic key pair is more difficult to hack than passwords. Keys are at least 1024 bits (128 bytes) long and the longer the key length, the longer it takes for an attacker to hack the keys.
- Keys are generated using an algorithm, which are not predictable and making them more secure than user-generated passwords.
- Only the device where the private key is stored can be used for remote access.
- The use of the public and private key pair ensures that when one key is compromised, the other remains secure.

This activity assumes that you will be using PuTTY for the PKA.

10.1. Server-side Key Storage

Step 1 Login to the administrator account of your virtual machine.

1a. Go to the `.ssh` hidden directory inside your home directory by using the command `cd ~/.ssh`. Does the `.ssh` directory exist?

Yes, it does exist.

1b. If the `.ssh` directory does not exist, create it inside your home directory. What command did you use to create the directory?

We would use the mkdir command if it did not exist.

1c. The `.ssh` directory should only allow the owner to have read, write, and execute access. Group members and the rest of the world should not be given any kind of access to this directory. What command should you use to apply this permission? Note: If you just created the .ssh directory, you need to apply this command to the directory.

jWe would use the command chmod 700 .ssh

```
drwx------  2 administrator administrator  4096 Feb  6 13:23 .ssh
-rw-r--r--  1 administrator administrator     0 Feb  6 13:43 .sudo_as_admin
administrator@lbysyad-s18-20162:~$
```

Step 2 Enter the command `ssh-keygen`. Leave the file name and passphrases blank. Simply press the enter key without typing anything when asked for these.

2a. Describe the output of the `ssh-keygen` command.

The output is a key's randomart image.

```
administrator@lbysyad-s18-20162:~/.ssh$ ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/home/administrator/.ssh/id_rsa):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/administrator/.ssh/id_rsa
Your public key has been saved in /home/administrator/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:X7HsHJSzL9DFl179q7wyGvq6A2ea3vKxnE2/Mmo5juY administrator@lbysyad-s18-20162
The key's randomart image is:
+---[RSA 3072]----+
|                 |
|             o ..|
|            = = o|
|           + O ..|
|        S . B . .|
|     . o . = o  .|
|      *..o. + .. |
|     =+=Booo...  |
|    +E=%B++o++.  |
+----[SHA256]-----+
administrator@lbysyad-s18-20162:~/.ssh$
```

2b. Check the contents of the `.ssh` directory. If the directory already exists at the start of the activity, are there new files inside the directory? If yes, what are these files? If you only created the directory for the activity, does the directory now have files inside? If yes, what are these files?

The files are 'authorized_keys', 'id_rsa', and 'id_rsa.pub'.

```
administrator@lbysyad-s18-20162:~/.ssh$ ls
authorized_keys
```

```
administrator@lbysyad-s18-20162:~/.ssh$ ls
authorized_keys  id_rsa  id_rsa.pub
```

2c. View the contents of the `id_rsa` file. What is the first and last line of this file?

The first line is 'BEGIN OPENSSH PRIVATE KEY', and the last line is 'END OPENSSH PRIVATE KEY'.

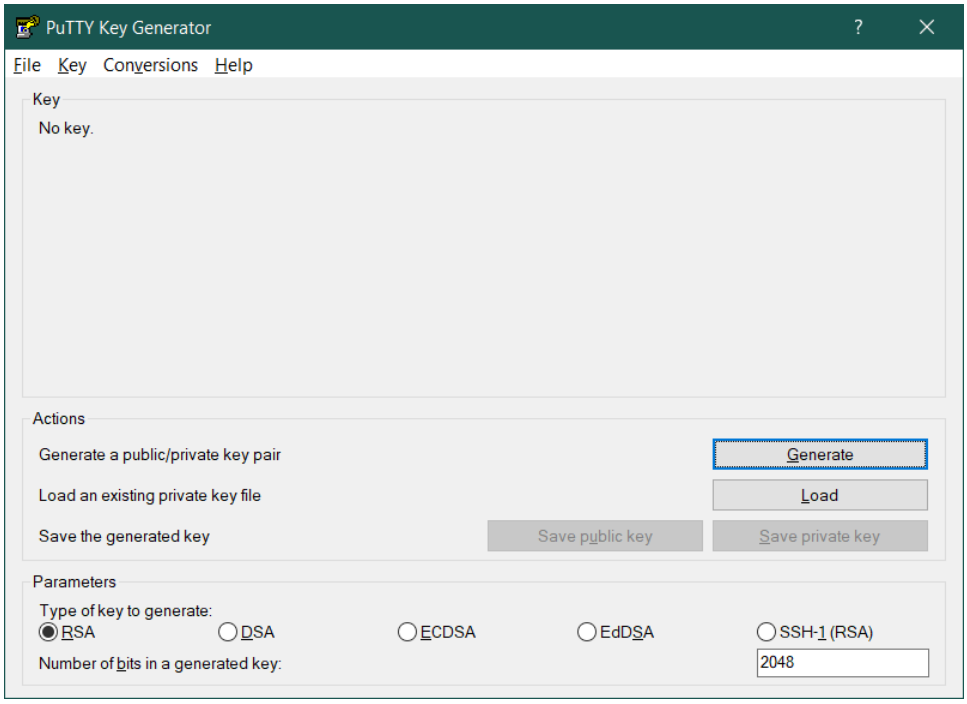
2d. View the contents of the `id_rsa.pub` file. Are the contents the same as the `id_rsa` file?

The contents of these two files do not match.

Since this activity is designed to generate keys for SSH login through PuTTY, these keys generated on the server side will not be used for now. Delete the `id_rsa` and `id_rsa.pub` files.

10.2. Creating New Public and Private Keys

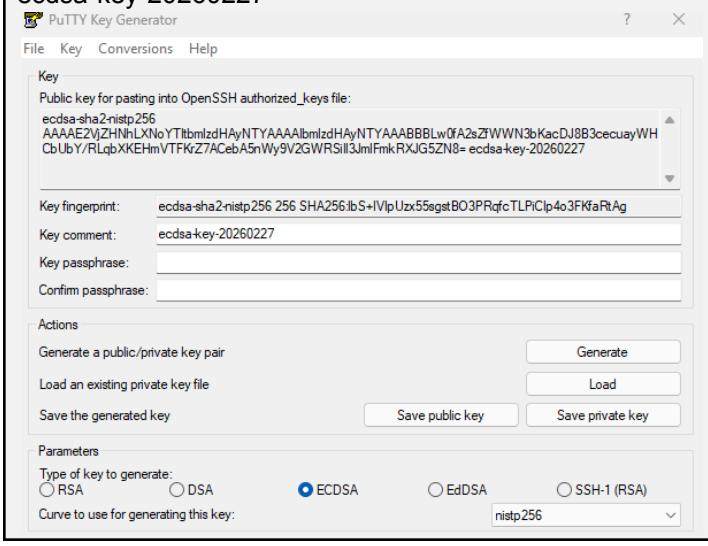
Step 3 On your machine, open the PuTTY Key Generator (PuTTYgen) software. If you do not have it yet, you can download it from the PuTTY website (https://www.chiark.greenend.org.uk/~sgtatham/putty/latest.html, look for puttygen.exe).



3a. Click on the **Key** menu then **Parameters** for saving key files…. Select **2** as the **PPK file version**.

3b. On the Parameters section, make sure that **ECDSA** is selected as type of key and curve is **nistp256**. Click **Generate**. Move your cursor over the blank area inside the **Key** section to add randomness to the keys that will be generated. Once the keys have been generated, the blank area will have content. Describe the contents now of the **Key** section.

This is the content of the key section 'ecdsa-sha2-nistp256
AAAAE2VjZHNhLXNoYTItbmlzdHAyNTYAAAAIbmlzdHAyNTYAAABBBLw0fA2sZfWWN3bKacDJ8B3cecuayWHCbUbY/RLqbXKEHmVTFKrZ7ACebA5nWy9V2GWRSiII3JmlFmkRXJG5ZN8=
ecdsa-key-20260227'



3c. Click **Save Private Key**. Click **Yes** when asked if you do not want to use a passphrase. Save the **.ppk** file to any accessible directory in your computer. Use the filename **id_ecdsa.ppk**.

3d. You may also save the public key or just keep the PuTTYgen window open. You will need the value of the public key for setting up the server. If you accidentally close the window, you can still retrieve the public key

by reopening PuTTYgen, selecting the Conversions menu option then Import key. From there, select your private key file (.ppk).

10.3. Copying the Public Key to the Server

Step 4 Login to the administrator account of your virtual machine using PuTTY.

4a. Create a file called `authorized_keys` inside your `.ssh` directory. What command did you use to create the file?

```
touch authorized_keys
```

4b. Copy the contents of your public key to the `authorized_keys` file. To do this, right click on the PuTTY interface to paste the contents of your clipboard. Make sure that there is only a single line that starts with `ecdsa-sha2-nistp256` and ends with `ecdsa-key-<YYYYMMDD>` where `YYYYMMDD` is the date when you created the key pair. Save the file after editing.

4c. Change the permissions of the `authorized_keys` file to reflect the following permissions: owner has read and write access, group members have read-only access, and the rest of the world has read-only access. What command did you use to change the permission?

```
chmod 644 authorized_keys
```

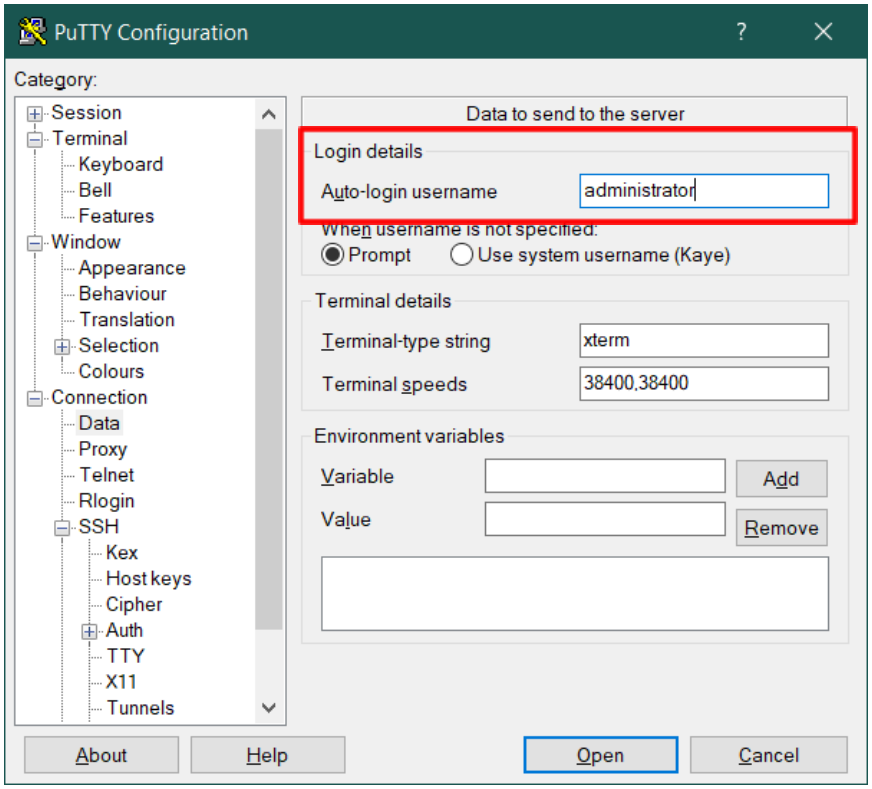
4d. Restart the SSH service using `sudo systemctl restart ssh`.

4e. Log out of your virtual machine.

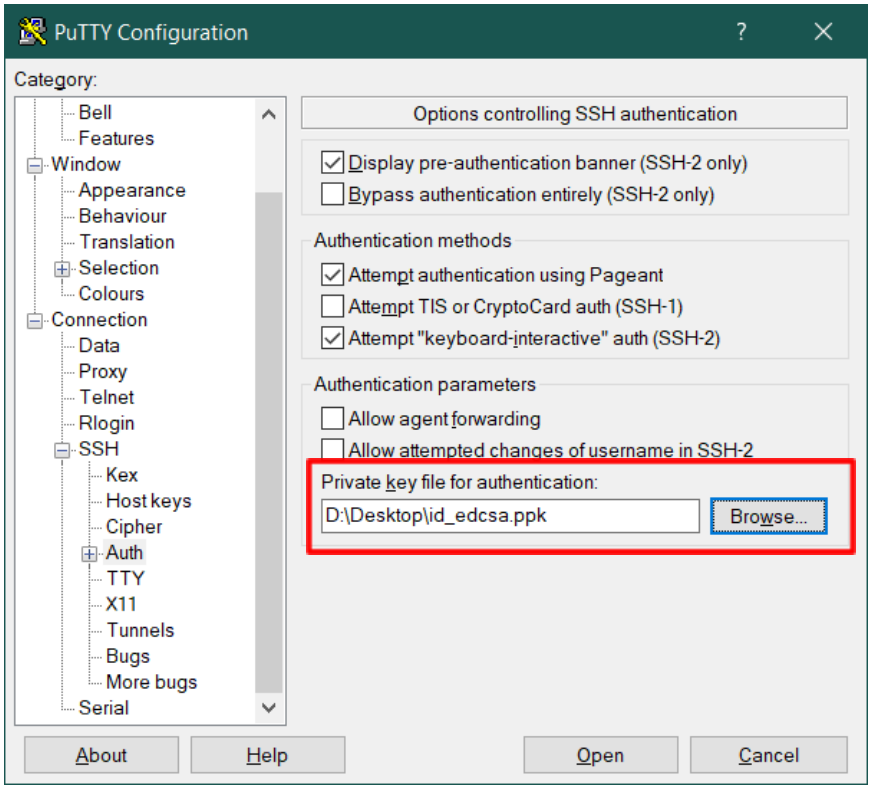
10.4. Connecting to the Server Using the Private Key

Step 5 Open PuTTY again but do not connect yet to the virtual machine. If you have a saved session, you may load it but do not open a connection for now. If you do not have a saved session, input the host name and port number manually.

5a. Under **Connection** > **Data**, enter *administrator* in the **Auto-login username** field. If your virtual machine username is not *administrator*, use the appropriate username instead.



5b. Under **Connection** > **SSH** > **Auth**, select the private key that you generated earlier under the **Private key file for authentication** section. Note: In other versions of PuTTY, the private key is found under **Connection > SSH > Auth > Credentials**. The screenshot may not be similar to the version you are using.



5c. Proceed with opening an SSH connection to your virtual machine. Were you able to login to your server?

Yes!

```
administrator@lbysyad-s18-20162: ~
Using username "administrator".
Authenticating with public key "ecdsa-key-20260227"
Welcome to Ubuntu 22.04.3 LTS (GNU/Linux 5.15.0-170-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage

  System information as of Sun Mar  1 12:00:17 AM PST 2026

  System load:  0.0                Processes:              107
  Usage of /:   34.6% of 23.45GB   Users logged in:        1
  Memory usage: 5%                 IPv4 address for ens18: 10.2.9.63
  Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

99 updates can be applied immediately.
To see these additional updates run: apt list --upgradable
```

5d. During the authentication process, were you asked for a password?

No. It was already logged in.

5e. What was the response of the server before the `Welcome to Ubuntu...` message?

```
administrator@lbysyad-s18-20162: ~
Using username "administrator".
Authenticating with public key "ecdsa-key-20260227"
```

10.5. Conclusion/Reflection

How is PKA possible given that the key pair have different values (one public and one private, instead of the same private key)?

The public and private key are mathematically linked cryptographic keys. Wherein the public key is shared openly to encrypt data or verify signatures, while the corresponding private key is kept secret to decrypt data or sign transactions

Why is the public key stored in the server and the private key stored in the client machine?

This process is to ensure a secure and an authentication without password while also making sure that only the owner can make a connection. The public key (server) acts like an open lock while the private key (client) is used as a secret key.

Is it technically possible for several users (using different devices) to use the same key pair for authentication by the same server? Why or why not?

Yes, a private key can be copied to multiple devices. However it is a risk to do so as you may compromise the users. There will be a loss of accountability. If multiple people use the shared key, it will be difficult to see who performed a certain command as logs mostly use the key for identifying only.

The advantages of the use of PKA over passwords were already mentioned in Page 1 of this lab activity. What are its disadvantages if there are any?

One disadvantage is that the private key is usually stored locally on the hard drive. So, if the hard drive is corrupted the user can be permanently locked out. Moreover, if the device gets stolen the thief can gain access to the private key. Another disadvantage is that it can be difficult to manage. Especially, when handling numerous devices. Manually updating them is tedious and prone to human-error.